

EMERALD

Performance Evaluation Analysis

Intelligent Traffic Violation Detection & Enforcement System

1.1 Vehicle Detection — YOLOv8 Benchmarks

Multiple independent studies establish YOLOv8's performance range across vehicle detection tasks under real-world traffic conditions.

Study / Source	Model Variant	Precision	Recall	mAP@50	FPS
Multi-class vehicle detection (5-fold cross-validation)	YOLOv8m	0.932	0.883	0.947	—
Variable lighting conditions (urban intersections)	YOLOv8	0.885	0.770	~0.85	—
Enhanced YOLOv8 (drone UAV) VisDrone2019 dataset	YOLOv8 improved	—	—	↑ 1.18% over baseline	—
Real-time inference speed benchmark	YOLOv8n	—	—	~0.88–0.91	±8.77 ms/frame



Key Finding: YOLOv8n (nano) prioritises speed over accuracy. Published ceiling is mAP@50 ~0.88–0.91 under clean conditions, dropping to 0.77–0.85 under variable lighting — precisely the condition at Indian urban intersections.

1.2 Helmet Detection & 1.3 Multi-Object Tracking

Helmet Violation Detection

Helmet detection on two-wheelers is a well-studied problem. The following benchmarks represent the state of the art with custom-trained YOLOv8 models on dedicated helmet datasets.

Study / Source	Model	Precision	mAP@50
Enhanced YOLOv8 + Sparse Spatial Attention (helmet-on)	YOLOv8 enhanced	93.49%	—
Enhanced YOLOv8 (helmet-off / no helmet)	YOLOv8 enhanced	92.68%	—
Helmet + Mirror + Plate System (overall pipeline)	YOLOv8	91.47%	0.843
YOLOv8-CGS construction helmet (SHD dataset)	YOLOv8-CGS	94.58%	93.18%

 **Key Finding:** Published precision range for helmet detection with custom-trained YOLOv8: 91–94%. EMERALD runs the base COCO model — numbers will sit below this range, which is expected. The multi-frame consensus mechanism compensates by suppressing single-frame false positives.

Multi-Object Tracking — ByteTrack

ByteTrack is the published state-of-the-art for multi-object tracking. Its benchmark numbers on MOT17 establish the reference ceiling.

Dataset / Benchmark	Tracker	MOTA	IDF1	FPS
MOT17 test set (primary benchmark)	ByteTrack	80.3	77.3	30
MOT17 + STRAM enhancement	ByteTrack + STRAM	81.0	79.9	—
MOT20 (denser, harder scenes)	ByteTrack	77.4	75.2	—
HiEve (complex events)	ByteTrack	61.7	63.1	—
BDD100K (multi-class traffic)	ByteTrack	↑ 1.5% vs ReID	↑ 0.8% vs ReID	—

 **Key Finding:** IDF1 is the most relevant metric for EMERALD — it measures identity preservation across frames, which directly determines whether the violation engine tracks the correct vehicle through a consensus window. MOTA captures detection-level accuracy. Both are needed.

1.4 OCR — Indian Number Plate Recognition

EasyOCR is the most widely used open-source OCR engine for Indian number plates. Published benchmarks across multiple systems establish expected accuracy ranges.

Study / Source	Detection Model	OCR Engine	Accuracy	Inference Time
Indian ANPR pipeline (YOLOv11 + EasyOCR)	YOLOv11	EasyOCR	88.2% char-level	~43 ms/image
EasyOCR vs Tesseract comparative study	YOLOv5	EasyOCR	>95%	—
EasyOCR vs Tesseract comparative study	YOLOv5	Tesseract	~90%	—
Indian ANPR (Springer) YOLOv5 + EasyOCR	YOLOv5	EasyOCR	99.5% precision, 98.8% recall	—
Real-time plate system YOLOv7x + EasyOCR	YOLOv7x	EasyOCR	~99%	real-time

📌 **Key Finding:** EasyOCR on Indian plates achieves 88–95% character-level accuracy under standard conditions. Accuracy drops with motion blur, oblique angles, and low light — all common in Bengaluru intersection footage. Character-level Levenshtein ratio is the correct metric, not whole-plate exact match.

2.1 Detection & 2.2 Tracking Strategy

2.1 Why YOLOv8n — and How We Compensate

YOLOv8n is chosen for speed — the pipeline must run in real-time on standard hardware without GPU optimisation. The published accuracy trade-off versus YOLOv8m or YOLOv8l is known and accepted. EMERALD compensates through two mechanisms that don't exist in published baseline implementations:

- **Asymmetric thresholding** — Per-class confidence floors rather than a single global threshold. Two-wheelers and large vehicles have fundamentally different detection difficulty profiles. A global threshold either over-detects large vehicles or under-detects motorcycles.
- **Dual-pass fallback** — When the primary pass finds zero two-wheelers in a frame, a second lower-confidence cropped scan is triggered. This recovers the recall gap on small objects near frame edges — the exact failure mode documented in YOLOv8n benchmarks under dense urban conditions.

Expected Performance Range: Detection mAP@50 0.82–0.88 (baseline YOLOv8n range). With dual-pass fallback, two-wheeler recall expected to recover 4–8 percentage points toward the YOLOv8m ceiling of 0.947.

2.2 Dual ByteTrack — Why One Config Cannot Serve Both Vehicle Classes

Published ByteTrack benchmarks (MOTA 80.3, IDF1 77.3) are measured on MOT17 — a pedestrian-dominant benchmark with relatively uniform motion signatures. Urban traffic presents two fundamentally different motion profiles simultaneously:

- **Motorcycles** — Fast two-wheelers near the camera: high velocity, frequent occlusion, small bounding box. Require low lost-track buffer and aggressive re-identification.
- **Trucks / buses** — Stationary or slow large vehicles at distance: stable position, large bounding box, minimal velocity. Require high lost-track tolerance to avoid fragmented tracks at lights.

A single ByteTrack config optimised for one class degrades on the other. EMERALD runs two independently tuned ByteTrack instances with separate activation thresholds, lost-track buffers, and matching tolerances. The key evaluation metric is IDF1 — identity switches per class — not aggregate MOTA.

Expected Performance Range: IDF1 target: >70 per class (vs ByteTrack baseline 77.3 on easier MOT17 dataset). Mixed urban traffic with varied lighting is harder — 68–74 IDF1 per class is a realistic and credible target.

2.3 Violation Engine & 2.4 OCR Strategy

2.3 Multi-Frame Consensus — Why Single-Frame Detection Is Insufficient for Enforcement

Published helmet detection systems achieve 91–94% precision on static images or single-frame crops. In a live video pipeline, single-frame detection produces a false positive rate that is unacceptable for enforcement — a momentary occlusion, a detection artefact, or a rapid pose change can trigger an incorrect event.

EMERALD's violation engine applies multi-frame consensus voting to three violation types where this matters most:

- **Wrong-side driving** — A vehicle must be confirmed travelling against flow across several consecutive frames before a violation is recorded. Single-frame wrong-way events — from lane-straddling, turning, or detection noise — are filtered.
- **Illegal parking** — A vehicle must remain stationary beyond a configurable duration window. Momentary stops at intersections are excluded.
- **Helmet non-compliance** — Helmet state must be consistent across multiple frames of the same tracked vehicle. The consensus threshold is configurable per deployment.

Expected Performance Range: Consensus mechanism trades recall for precision — effective precision on consensus-gated violations expected to reach 92–96%, above the single-frame published ceiling, at a small cost to recall (2–4 percentage points). This is the correct trade-off for an enforcement system.

2.4 Lazy Plate Recognition — Why We Don't Run OCR on Every Frame

Published ANPR systems benchmark OCR accuracy on clean, dedicated plate crops with no frame-rate constraint. In a live multi-vehicle pipeline, running EasyOCR on every vehicle in every frame is computationally prohibitive and unnecessary.

EMERALD's OCR runs lazily — exclusively on vehicles that have already crossed a violation threshold. This means:

- **Efficiency** — Compute load is bounded by violation frequency, not traffic volume. A busy intersection with 200 vehicles per minute but 3 violations per minute runs OCR 3 times per minute.
- **Accuracy relevance** — OCR is applied only to violation-confirmed vehicles, where the plate read serves a definite enforcement purpose. Accuracy on this subset is the relevant measure.
- **Format validation** — The regex pattern for Indian plate formatting (AA00AA0000) acts as a post-OCR validation filter, rejecting misreads that don't conform to known plate formats.

Expected Performance Range: Character-level accuracy target: 85–92% (EasyOCR published range for Indian plates under real conditions). Exact whole-plate match rate expected lower: 70–80%, reflecting the harder multi-character recognition problem under variable lighting.

3.1 Component Status Overview

Every component described in Parts 1 and 2 mapped to its current status in the working prototype.

Component	Case Study Benchmark	EMERALD Approach	Prototype Status
Vehicle Detection (YOLOv8n)	mAP@50: 0.88–0.94 (YOLOv8m, custom dataset)	YOLOv8n with per-class thresholds + dual-pass motorcycle fallback	✔ Implemented
Multi-Object Tracking (ByteTrack)	MOTA 80.3, IDf1 77.3 (MOT17, single config)	Dual ByteTrack instances tuned per vehicle class (separate thresholds)	✔ Implemented
Helmet Detection	Precision 91–94% (custom-trained models)	Base COCO model + multi-frame consensus voting (3+ frames)	✔ Implemented
Stop-Line / Red-Light Detection	No direct benchmark (rule-based geometry)	Zone-based rule engine with HSV traffic-light colour classifier	✔ Implemented
Wrong-Side Detection	No published benchmark (unsolved in deployment)	Multi-frame direction consensus voting across consecutive frames	✔ Implemented
Illegal Parking Detection	No published benchmark (unsolved in deployment)	Stationary-duration window with configurable threshold	✔ Implemented
OCR / Plate Reading (EasyOCR)	88–95% char-level (Indian plates, clean)	Lazy OCR on violation-confirmed vehicles only + regex validation	✔ Implemented (unbenchmarked)
Triple-Riding Detection	No published benchmark	Rider-count logic in rule engine (data pipe not yet connected)	⚙ Framework Designed
Seatbelt Detection	Hard small-object problem (YOLOv5l/x + CNN fallback proposed in literature)	Binary classifier on cropped windshield region (Phase 2)	⚙ Framework Designed
Repeat-Offender Ledger	No equivalent in deployed systems	Persistent plate-keyed ledger with fuzzy match (edit-distance 1)	⚙ Framework Designed
Evidence Generation (Frame + JSON)	Standard in published ANPR/violation systems	Annotated JPEG + JSON-Lines record at event time	✔ Implemented
Violation Analytics & Reporting	Batch reporting in deployed systems	Real-time filtered query + trend plots + HTML export	✔ Implemented
Performance Evaluation Harness	Standard benchmark scaffolds in literature	FPS, mAP, precision, recall, F1 scaffold built and runnable	⚙ Framework Designed (no ground-truth run)